

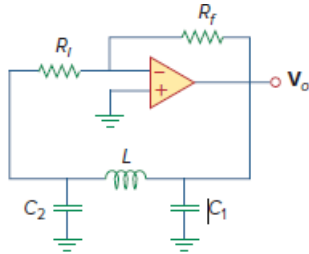
Problem

Figure 10.135 shows a *Colpitts oscillator*. Show that the oscillation frequency is

$$f_o = \frac{1}{2\pi\sqrt{LC_T}}$$

where $C_T = C_1C_2/(C_1 + C_2)$. Assume $R_i \gg X_{C_2}$.

Figure 10.135: A Colpitts oscillator



Step-by-step solution

Step 1 of 5

Draw the following circuit diagram:

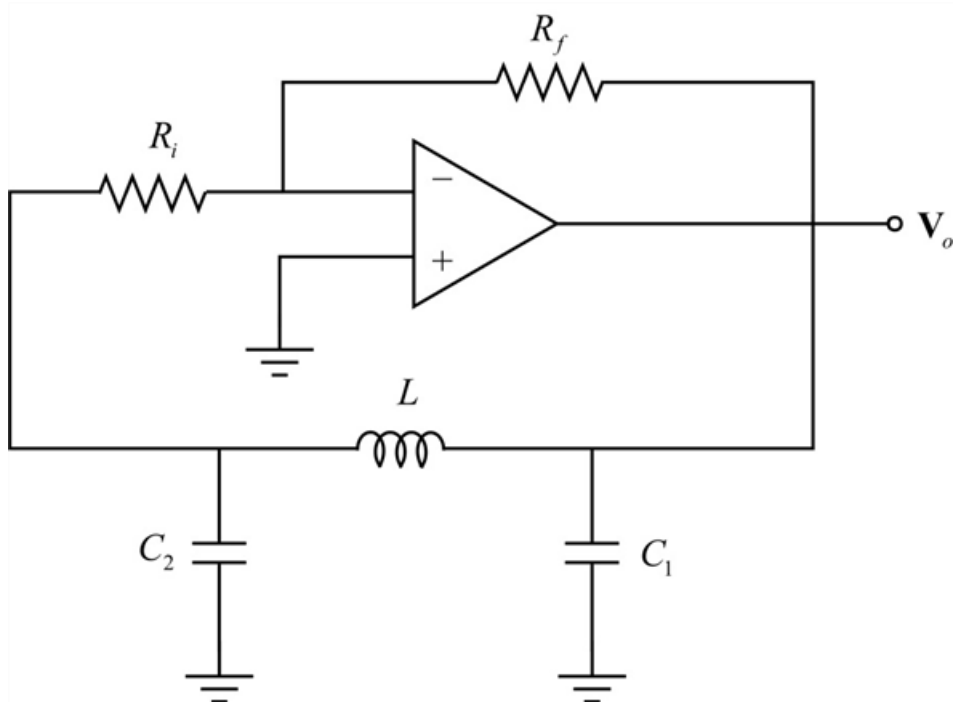


Figure 1

Step 2 of 5

Draw the feedback part of the circuit.

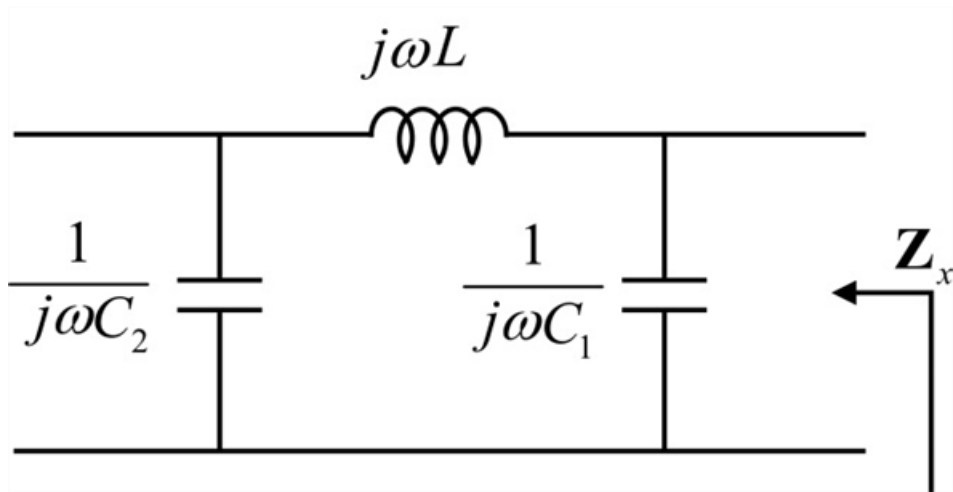


Figure 2

Step 3 of 5

Calculate the impedance $\mathbf{Z}_x$ for the circuit in Figure 2.

$$\begin{aligned}\mathbf{Z}_x &= \left[\frac{1}{j\omega C_2} + j\omega L \right] \parallel \left(\frac{1}{j\omega C_1} \right) \\ &= \left[\frac{-j}{\omega C_2} + j\omega L \right] \parallel \left(\frac{-j}{\omega C_1} \right) \\ &= \frac{\left[\frac{-j}{\omega C_2} + j\omega L \right] \left(\frac{-j}{\omega C_1} \right)}{\left[\frac{-j}{\omega C_2} + j\omega L \right] + \left(\frac{-j}{\omega C_1} \right)} \\ &= \frac{\left[\frac{-1}{\omega^2 C_1 C_2} + \frac{L}{C_1} \right]}{j \left[\frac{-1}{\omega C_2} + \omega L - \frac{1}{\omega C_1} \right]}\end{aligned}$$

Step 4 of 5

Simplify $\mathbf{Z}_x$ further:

$$\begin{aligned}\mathbf{Z}_x &= \frac{\left[\frac{\omega^2 L C_2 - 1}{\omega^2 C_1 C_2} \right]}{j \left[\frac{-C_1 + \omega^2 L C_2 - C_2}{\omega C_1 C_2} \right]} \\ &= \frac{\left[\omega L C_2 - \frac{1}{\omega} \right]}{j \left[-C_1 + \omega^2 L C_2 - C_2 \right]} \\ &= \frac{\left[\frac{1}{\omega} - \omega L C_2 \right]}{j \left[C_1 + C_2 - \omega^2 L C_1 C_2 \right]}\end{aligned}$$

Step 5 of 5

The impedance $\mathbf{Z}_x$ is purely real, when the imaginary part equal to zero.

$$C_1 + C_2 - \omega_o^2 LC_1 C_2 = 0$$

$$\omega_o^2 LC_1 C_2 = C_1 + C_2$$

$$\omega_o^2 = \frac{C_1 + C_2}{LC_1 C_2}$$

$$(2\pi f_o)^2 = \frac{1}{L \left(\frac{C_1 C_2}{C_1 + C_2} \right)}$$

$$2\pi f_o = \frac{1}{\sqrt{LC_T}} \quad \left(\text{where, } C_T = \frac{C_1 C_2}{C_1 + C_2} \right)$$

$$f_o = \frac{1}{2\pi \sqrt{LC_T}}$$

Therefore, the frequency $f_o = \frac{1}{2\pi \sqrt{LC_T}}$ where $C_T = \frac{C_1 C_2}{C_1 + C_2}$ is proved.